

You are evaluating ontology reuse and namespace ownership. This is a non-interactive task. Use the GPT 5.5 High Model. Return exactly one valid downloadable JSON file and nothing else.

Do not:

- ask questions or request clarification;
- output Markdown, comments, or code fences;
- explain your reasoning outside the required JSON fields;
- summarize the ontology outside the schema.

Task

Analyze how FASTO reuses external ontologies and namespaces. Identify:

- all reused ontologies and external ontology namespaces supported by the evidence;
- the import or reuse mechanism used for each ontology;
- whether reused content is imported, copied, partially reused, indirectly imported, or reused through another mechanism;
- entities originating from reused ontologies;
- entities apparently created by FASTO inside a reused ontology namespace;
- reused entities that FASTO redeclares, annotates, constrains, or otherwise modifies;
- entities whose namespace ownership is ambiguous, suspicious, or inconsistent;
- reuse, provenance, import, namespace, versioning, and maintainability issues.

Pay particular attention to FASTO-specific classes, properties, individuals, or axioms that use a reused ontology namespace instead of FASTO's namespace.

Evidence

Four evidence sources are provided (7 files).

1. Primary evidence: The original FASTO_RDF.owl in RDF/XML syntax. This is the authoritative source.
2. Secondary evidence: A rule-based verbalizer generated from a temporary sanitized copy of FASTO. This exists only to improve readability.
3. Supplementary evidence: The paper describing FASTO.

Use the paper to clarify:

- which ontologies FASTO claims to reuse;
- which entities or concepts are described as FASTO-specific;
- whether the authors describe importing, adapting, copying, extending, or partially reusing external ontologies;
- intended provenance or ownership.

4. Additional .ttl and .owl files of the reused ontologies: FHIR, DMTO, OntoFood and SSN.

Evidence rules

- RDF/XML is always the primary source.
- The verbalizer is supplementary evidence only.
- Do not compare the RDF/XML and verbalizer.
- The paper must not override explicit RDF/XML evidence.
- If the paper and RDF/XML are inconsistent, report the inconsistency and treat the RDF/XML as authoritative for what is actually encoded.
- Report only conclusions supported by the supplied evidence.
- Do not invent source-ontology contents, ownership, versions, import histories, or intended semantics.

- The use of an external namespace alone does not prove direct copying or local creation.
- The absence of owl:imports alone does not prove direct copying.
- Distinguish directly reused, copied, locally created, and locally modified entities where the evidence permits.
- Use exact IRIs, namespaces, declarations, import statements, axioms, or paper statements as evidence.
- Use lower confidence and an unclear classification when ownership, reuse scope, or import mechanism cannot be determined reliably.
- Do not force a definitive classification from incomplete evidence.

Entity inventory rules

The classes, object_properties, data_properties, and individuals arrays under directly_reused_entities, local_entities_in_reused_namespace, and entities_with_unexpected_namespace are exhaustive entity inventories.

For every such array:

- Include every qualifying entity supported by the evidence.
- Each array element must be exactly one complete entity IRI.
- Do not combine multiple IRIs in one element.
- Do not include counts, descriptions, evidence, source locations, qualifications, or explanatory prose.
- Do not replace complete inventories with representative examples.
- Do not use phrases such as "including", "such as", "among others", "related entities", "entity set", or "N entities".
- Do not use ellipses, "etc.", placeholders, or omission statements.
- Preserve each IRI exactly as encoded.
- Remove exact duplicates.
- The same entity may appear in more than one category when multiple category definitions apply.

Axiom array rules

- Use one distinct axiom or one precisely defined homogeneous axiom pattern per element.
- Identify the affected entity IRI, axiom type or predicate, and object or restriction when available.
- Do not replace identified axioms with counts and selected examples.
- Group an axiom pattern only when the same reasoning applies and all affected entities are already individually enumerated in the applicable entity arrays.

Issue grouping rules

The exhaustive-enumeration requirement does not require one detected issue per entity.

Entities may be grouped only inside detected_issues, and only when they share the same:

- issue type;
- namespace role;
- provenance assessment;
- modification status;
- severity;
- evidential reasoning;
- explanation;
- suggested fix;
- confidence level.

For a grouped issue:

- describe the entity set precisely in the entity field;
 - do not repeat the complete IRI inventory in the issue;
 - ensure every affected entity is individually listed in the applicable entity arrays.^[SEP]
- Create separate issues when entities differ materially in provenance, modification status, entity type, namespace role, issue type, severity, evidence, explanation, suggested fix, or confidence.

Reporting rules

Report entity-level issues separately when entities differ in:

- provenance;
- modification status;
- namespace role;
- issue type;
- supporting evidence.

Entities may be grouped only inside `detected_issues` when the issue-grouping rules are satisfied. Entities must never be grouped, counted, summarized, or represented by examples in the entity inventory arrays. Do not merge findings merely because entities share a namespace. Remove exact or substantively equivalent duplicate issues.

Output rules

- Return exactly one JSON object conforming to the schema.
- Do not create fields outside the schema.
- Include all required objects and arrays, even when empty.
- Escape all evidence as valid JSON strings.
- Use confidence values from 0.0 to 1.0.
- Use globally unique sequential issue IDs: RNS1, RNS2, RNS3, and so on.
- Do not restart issue numbering for each reused ontology.
- When `expected_original_namespace` cannot be established from the evidence, use an empty string and explain the uncertainty in the relevant assessment.

Field rules

`import_method`: This field is an array. Include one or more of:

- `complete_import`
- `partial_import`
- `indirect_import`
- `direct_copy`
- `no_import_detected`
- `unclear`

Use multiple values only when the evidence supports multiple reuse mechanisms for the same ontology. `reuse_scope`: Select exactly one:

- `complete`
- `partial`
- `unclear`

`directly_reused_entities`: List entities or axioms supported as originating from or being

directly reused from an external ontology. `local_entities_in_reused_namespace`: List entities or axioms supported as being created by FASTO while using a reused ontology namespace.

`entities_with_unexpected_namespace`: List entities whose namespace appears inconsistent with their apparent ownership, declaration context, or description in the paper.

detected_issues: This field is an array of issue objects.

For each issue, select exactly one issue_type:

- copied_reuse
- namespace_extension
- namespace_mismatch
- missing_import
- partial_reuse_without_provenance
- local_modification_of_reused_entity
- unexpected_external_namespace
- unclear_ownership
- version_ambiguity
- other

Use other only when none of the listed values applies, and explain the issue in additional_comment. For each issue, select exactly one entity_type:

- class
- object_property
- data_property
- individual
- axiom
- ontology
- unknown

For each issue, select exactly one severity:

- error
- warning
- info

The summary must synthesize only evidence and issues already represented in the corresponding ontology assessment. It must not introduce new conclusions.

Required JSON schema

```
{
  "task": "reuse_namespace_analysis",
  "model": "",
  "input_condition": "llm_only",
  "ontology": {
    "name": "FASTO",
    "base_namespace": "",
    "analyzed_reused_ontologies": []
  },
  "reused_ontology_assessments": [
    {
      "reused_ontology": "",
      "expected_original_namespace": "",
      "detected_namespaces": [],
      "import_method": [],
      "reuse_scope": "unclear",
      "directly_reused_entities": {
        "classes": [],
        "object_properties": [],
        "data_properties": [],

```

```

"individuals": [],
"axioms": []
},
"local_entities_in_reused_namespace": {
"classes": [],
"object_properties": [],
"data_properties": [],
"individuals": [],
"axioms": []
},
"entities_with_unexpected_namespace": {
"classes": [],
"object_properties": [],
"data_properties": [],
"individuals": [],
"axioms": []
},
"detected_issues": [
{
"id": "RNS1",
"entity": "",
"entity_type": "unknown",
"issue_type": "unclear_ownership",
"severity": "warning",
"evidence": "",
"explanation": "",
"suggested_fix": "",
"confidence": 0.0,
"additional_comment": ""
}
],
"summary": {
"main_problem": "",
"maintenance_risk": "",
"recommended_action": ""
}
},
"global_comments": [
{
"comment": "",
"evidence": "",
"confidence": 0.0
}
]
}

```

Internal quality checks

Before returning the JSON, internally verify that:

- every reused ontology and namespace is supported by evidence;

- every ownership or reuse claim has supporting evidence;
- direct copying was not inferred only from missing owl:imports;
- uncertain cases use appropriate confidence and classification;
- entity-level differences were not incorrectly merged;
- exact or substantively equivalent duplicates were removed;
- every qualifying entity is listed as a separate complete IRI;
- no entity-array element contains prose, a count, or multiple IRIs;
- every exact source-ontology match is included in `directly_reused_entities`;
- every FASTO declaration in a reused namespace was considered for local or unclear ownership;
- every reused entity receiving FASTO-local axioms was considered for `local_modification_of_reused_entity`;
- every grouped issue is supported by complete entity enumeration;
- no placeholder, ellipsis, truncation notice, or omitted-remainder statement occurs;
- all required fields and arrays are present;
- issue IDs are unique, global, and sequential;
- the final output is valid JSON.

Now analyze the supplied evidence and return the downloadable JSON file only.